SELVAMANI S

Software Engineer (.NET) | Backend & Full Stack

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SUMMARY

.NET Software Engineer with **3.5+ years** of experience architecting and modernizing public-sector e-governance and revenue ERP systems serving **690K+ citizen and consumer records** in **C#, ASP.NET Core, SQL Server, and PostgreSQL**. Cut primary database storage by **70%** via MinIO object storage, led framework migrations from **.NET Core 3.1 to .NET 8** and **.NET 8 to .NET 10**, and integrated national payment gateways (**BBPS, GRAS**) with automated background reconciliation. Proven expertise in Clean Architecture, REPR pattern, and compile-time CQRS.

TECHNICAL SKILLS

Languages & Frameworks: C#, .NET 10 / 8 / Core 3.1, ASP.NET Core (MVC, REST Web API), FastEndpoints, SignalR, LINQ, SQL, JavaScript

Architecture & Patterns: Clean Architecture, SOLID, REPR Pattern, CQRS (Source-Generated Mediator), Specification Pattern, Dependency Injection, Refit, Mapster, .NET Aspire, Coravel

Databases & Storage: MS SQL Server, PostgreSQL, Entity Framework Core (EF Core), ADO.NET, Stored Procedures, MinIO (S3 Object Storage)

Reporting, Security & Observability: QuestPDF, ClosedXML, EPPlus, PDF.js, FIDO2 / WebAuthn, JWT, 2FA / OTP, OpenTelemetry, Serilog

Frontend & DevOps: HTMX, Alpine.js, Bootstrap 5, Chart.js, DataTables, Unit Testing (MSTest), Git, GitLab CI/CD

PROFESSIONAL EXPERIENCE

Full Stack Developer
Nov 2024 – Present

AeoLogic Technologies (Client: NICSI / Puducherry Government | Municipal WaterTax ERP) — Puducherry, India

- **Cut SQL Server storage footprint by 70%** by offloading consumer document attachments to self-hosted **MinIO (S3)** object storage with non-blocking, on-the-fly streaming ZIP compression.
- **Integrated BBPS (JWT-secured bill fetch/pay), GRAS, and LGRAMS payment gateways**; engineered idempotent background reconciliation jobs via **Coravel** to automatically resolve dropped transactions without manual ledger intervention.
- **Engineered core billing engines** (tariff slabs, arrears, surcharges), new-connection approval workflows, and counter-collection modules for **221K+ municipal water consumers** using **Ardalis Clean Architecture** and **FastEndpoints**; upgraded the platform from **.NET 8 to .NET 10**.
- **Eliminated runtime reflection overhead** and avoided commercial licensing lock-in by migrating command/query pipelines from **MediatR** to a compile-time, **source-generated Mediator**.
- **Built real-time cashier notifications** and instant receipt triggers via **SignalR**, and redesigned statutory demand notices, receipts, and bills using **QuestPDF**.
- **Hardened application security** with passwordless **FIDO2 passkeys**, OTP-based 2FA, and SQL-backed distributed sessions enforcing strict active-session concurrency limits.
- **Streamlined inter-service HTTP communication** using type-safe **Refit** clients and configured **.NET Aspire** for local multi-service orchestration with unified OpenTelemetry tracing and health checks.

Software Developer
May 2023 – Oct 2024

SISL Infotech (Client: NICSI / Labour Dept., Puducherry | Employment Exchange Platform) — Puducherry, India

- **Delivered core citizen e-governance workflows** (online registration, 3-year renewal, regional transfer, and vacancy sponsorship) across **4 regional exchanges** for **472K+ registered job seekers (134K+ active live profiles)** using **ASP.NET Core MVC** and **PostgreSQL**.
- **Migrated a 430+ entity legacy database and codebase from .NET Core 3.1 to .NET 8**, refactoring complex **EF Core** LINQ queries and enforcing strict relational integrity across multi-column composite keys `(Exchange, Regyear, Regno)`.
- **Replaced proprietary DevExpress reporting with QuestPDF and ClosedXML** across 15+ statutory reports, eliminating commercial license costs and cutting server PDF memory overhead by **~40%**.
- **Engineered a zero-disk PDF streaming pipeline** that renders generated reports directly to an in-browser **PDF.js** viewer, eliminating temporary file storage and server disk I/O bottlenecks.
- **Cleared government security compliance audits (STQC / CERT-In)** by implementing session-bound dynamic salt challenge-response authentication and custom middleware enforcing a single active session per user.
- **Synchronized state candidate profiles** and automated transactional alerts with the National Career Service (NCS) portal and NIC SMS Gateway via **WCF/SOAP and REST APIs**.

Software Engineer Trainee
Apr 2022 – Apr 2023

Harktechs (B2B Seafood Export & Shipment Tracking Portal) — Puducherry, India

- **Developed B2B shipment tracking**, container allocation, and procurement workflows in **ASP.NET Web Forms** with AJAX partial-page updates.
- **Optimized SQL Server stored procedures** and **ADO.NET** data access layers for export invoice generation, inventory reconciliation, and vendor ledger reporting.

EDUCATION

Master of Computer Applications (MCA) | **Christ College of Engineering and Technology, Pondicherry University** — **2020 – 2022**

Bachelor of Computer Applications (BCA) | **Indira Gandhi College of Arts and Science, Pondicherry University** — **2017 – 2020**